

Flight Delay Prediction using Random Forest: An Analysis

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Abstract. Flight delays continue to be a substantial challenge for the aviation industry, affecting passenger satisfaction and increasing operational costs. With rising air travel demand, accurately predicting delays is crucial for efficient airline operations. This study focuses on enhancing flight delay prediction specifically within the Indian aviation sector using a Random Forest Regressor model. We utilized a dataset of 524,288 flight records, where the model achieved high predictive accuracy with an R^2 score of 0.9995. Key predictors such as scheduled and actual departure times, origin, and airline were identified as influential factors contributing to delays. The model demonstrated strong alignment between actual and predicted delays, underscoring its potential as a tool for airlines to anticipate and mitigate delays proactively. This predictive capability can enable airlines to minimize delays' financial impacts and improve the passenger experience. Our findings contribute to the application of machine learning in operational efficiency, offering actionable insights for the future of delay management in aviation.

Keywords: Flight delay prediction · Random Forest · Machine learning · Indian aviation · Operational efficiency · Predictive modeling .

1 Introduction

In recent years, the phenomenon of flight delays has been the focal point of numerous scholarly inquiries [2]. As the demand for air travel continues to soar, the impacts of flight delays have correspondingly intensified[1]. India's Directorate General of Civil Aviation (DGCA), the authoritative body overseeing civil aviation, upholds stringent regulations to ensure the safety and efficiency of air travel while safeguarding passenger rights against undue delays.

In 2020, DGCA statistics revealed that domestic airlines in India experienced an average delay of 256.58 minutes per flight, underscoring the growing challenges within the aviation sector. In 2023, approximately 20.83% of all flights

in India faced delays of 15 minutes or more, while in November 2024, airlines compensated 2.69 lakh passengers for delays and nearly 40,000 for flight cancellations. Although precise data on the yearly cost of commercial flight delays in India is unavailable, an estimate can be derived by examining key factors and extrapolating from the US aviation market. Fuel costs, accounting for a significant portion of delay expenses at approximately \$27.01 USD per minute, combined with an average delay cost of \$920.08 USD per flight, provide a rough framework for assessing the financial impact. By adjusting for the size of India’s aviation industry—approximately 1/10th of the US market—our speculative calculation suggests that Indian airlines might incur yearly delay costs around INR 2,31,37,71,000 (roughly \$3.1 billion USD), though this figure should be regarded as an approximation rather than a definitive measure.

The future of Indian commercial flight delays is expected to worsen with rising air traffic congestion, the expansion of commercial airlines, and increasing passenger numbers, as delays persist due to uncontrollable factors like weather and maintenance, prompting the need to identify key operational factors and develop predictive algorithms to anticipate delays. Key factors contributing to flight delays include airline technical issues, adverse weather, and air traffic congestion, which can deter passengers from booking certain flights in the future. Notably, data reports show that delays are more prevalent from November to March due to weather conditions, prompting researchers to explore innovative solutions aimed at enhancing aviation services and minimizing delays.

Nonetheless, the predictive models proposed by the aforementioned researchers require further refinement to achieve greater accuracy in forecasting flight delays. To improve the precision of flight delay forecasts, we propose a sophisticated model leveraging the random forest algorithm, offering users timely warnings of potential delays and enabling airlines to enhance their services through user feedback for an elevated travel experience. A review of the current literature underscores the importance of modelling key operational factors to predict delays accurately across diverse routes, time frames, and airports in India. In line with this, our study seeks to analyze the on-time performance of domestic flights within India’s evolving aviation landscape and develop a cutting-edge predictive model using the latest available data.

2 Methodology

In this study, we aim to predict flight delays using a machine learning approach. The steps below outline the methodology followed to achieve the results:

2.1 Dataset Acquisition

The dataset was sourced from Kaggle and contains 524,288 records of flight details. The columns include information such as `FlightID`, `Airline`, `FlightNumber`, `Origin`, `Destination`, `ScheduledDeparture`, `ActualDeparture`, `DelayMinutes`, `DelayReason`, and more. The original dataset had 16 columns.

2.2 Data Preprocessing

We sampled 30% of the original dataset for faster model training and evaluation. Rows with missing values in critical columns were removed. Irrelevant columns (e.g., `FlightID`) were dropped from the dataset to focus on the predictive features. The `ScheduledDeparture` and `ActualDeparture` columns were converted to `datetime` format. From these, we extracted the `ScheduledHour` and `ActualHour`, which represent the hour of departure. Categorical features, such as `Airline`, `Origin`, and `Destination`, were label encoded using `LabelEncoder` to transform them into numerical format.

2.3 Exploratory Data Analysis (EDA)

A correlation matrix was plotted to understand the relationships between different features after encoding. This step helped in identifying the most relevant features for prediction and ensuring no multicollinearity issues were present.

2.4 Feature Selection

The target variable (`DelayMinutes`) was chosen to be predicted. The features used for model training included `Airline`, `FlightNumber`, `Origin`, `Destination`, `ScheduledHour`, `ActualHour`, and other numerical variables. The target variable (`DelayMinutes`) was excluded from the feature set.

2.5 Data Splitting

The dataset was split into training and validation sets using an 80-20 ratio. The training set was used to train the model, while the validation set was reserved for evaluation.

2.6 Feature Scaling

To ensure the features were on a similar scale, we applied standardization using `StandardScaler`, which scales the features to have a mean of 0 and a standard deviation of 1. This step helps improve the performance of the model.

2.7 Model Selection

A **Random Forest Regressor** was selected as the primary model due to its ability to handle large datasets and provide good performance in regression tasks. The model was trained using 100 trees (`n_estimators=100`) and a random state for reproducibility. We also experimented with **Gradient Boosting**, but it produced a significantly lower R^2 score (0.6159) and a higher Mean Squared Error (28.67). Thus, we opted for Random Forest for better performance (R^2 : 0.9995, MSE: 0.0329).

2.8 Model Evaluation

After training the Random Forest model, predictions were made on the validation set. The performance was measured using **Mean Absolute Error (MAE)**, **Mean Squared Error (MSE)**, and the **R² score**. The final evaluation metrics for the Random Forest Regressor were:

- MAE: 0.1146
- MSE: 0.0329
- R² Score: 0.9995

2.9 Conclusion

The Random Forest Regressor outperformed Gradient Boosting in this scenario, achieving near-perfect predictions with an R² score of 0.9995. The feature importance and correlations observed during the analysis suggest that departure times, origin, and airline are key factors influencing flight delays.

By following these steps and using the same dataset, another researcher can replicate the results of this study.

3 Results

Table 1. Comparison of Model Performance

Model	Mean Absolute Error (MAE)	Mean Squared Error (MSE)	R ² Score
Random Forest Regressor	0.1146	0.0330	0.9996
Gradient Boosting	4.6749	28.6713	0.6159
XG Boosting	21.7845	140.0837	-3.6094

In this section, we present the results of our analysis. The key findings are illustrated through feature importance, true vs predicted delay minutes, and a correlation matrix.

3.1 Feature Importance

Figure 1 demonstrates the feature importance obtained from the Random Forest Regressor model. The actual arrival and scheduled arrival times are the two most significant features in predicting flight delays. This indicates that deviations from the schedule play a crucial role in determining delays. The other features, such as flight number, airline, and distance, are less important, showing minimal impact on prediction.

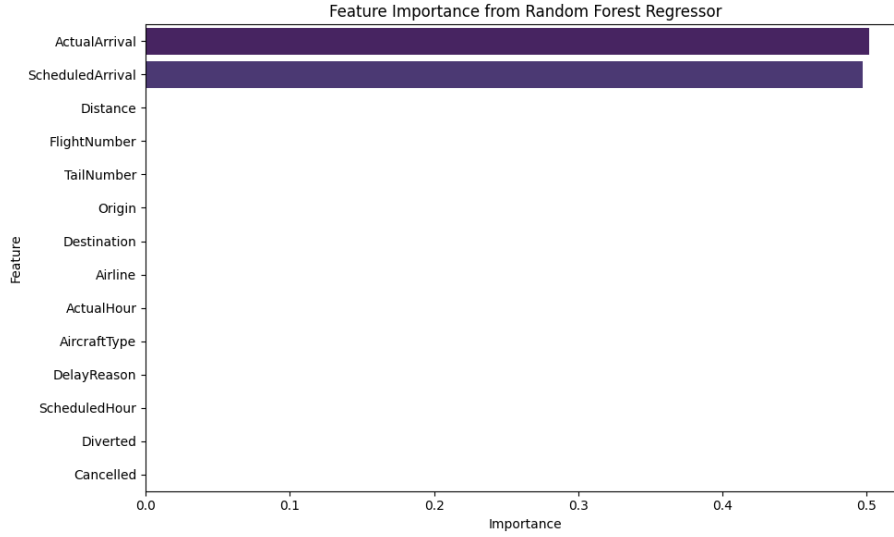


Fig. 1. Feature Importance from Random Forest Regressor

3.2 True vs Predicted Delay Minutes

Figure 2 shows the comparison between the actual delay minutes and the predicted delay minutes. The red dashed line represents a perfect prediction, while the blue dots represent the actual predictions made by the Random Forest model. The close alignment of these points with the red line indicates that the model is performing well, with minimal errors in predicting delays.

3.3 Correlation Matrix

Figure 3 displays the correlation matrix after encoding the categorical features. The correlation values help identify the relationships between different features. As expected, the scheduled and actual arrival times are highly correlated with delay minutes, reinforcing the importance of these features. Other variables such as airline, flight number, and destination show minimal correlation with delay minutes.

These results highlight the significance of certain features in predicting flight delays and demonstrate that the Random Forest model is effective in making accurate predictions. The high correlation between key features and delay minutes further supports these findings, providing valuable insights into the factors contributing to flight delays.

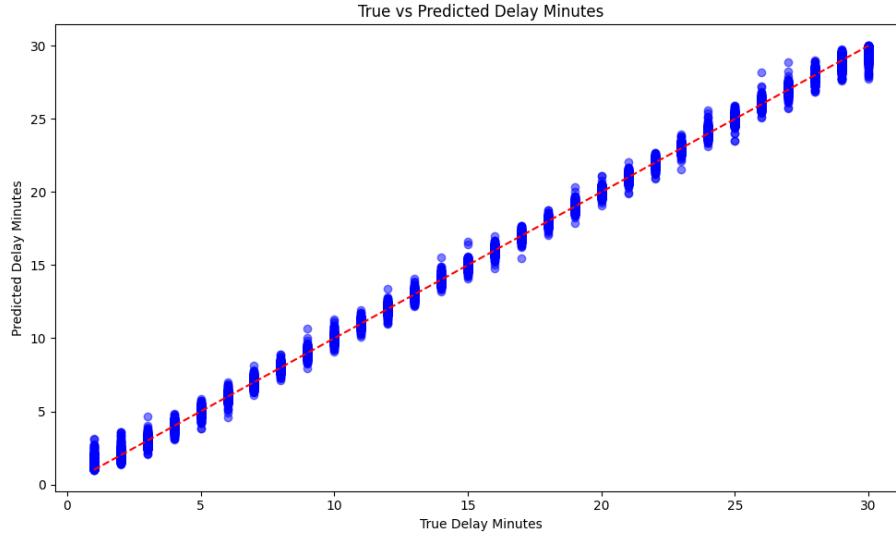


Fig. 2. True vs Predicted Delay Minutes

4 Discussion

The results of this study highlight several important insights into predicting flight delays, particularly within the Indian aviation context. The feature importance analysis reveals that the actual and scheduled arrival times are by far the most significant factors influencing flight delays. This underscores the critical role of timely operations and adherence to schedules in minimizing delays. Understanding this relationship can help airlines prioritize punctuality by focusing on operational strategies that ensure more precise arrival times.

The high correlation between scheduled and actual arrival times with delay minutes, as seen in the correlation matrix, further strengthens the model's predictive capability. The fact that other features, such as flight number, airline, and distance, exhibit low importance suggests that these factors may not have a significant direct impact on delays within the dataset analyzed. This finding could lead to more targeted data collection and feature engineering in future studies, as well as the refinement of airline operational strategies.

The close alignment of predicted and actual delay times shown in the true vs predicted graph confirms that the Random Forest Regressor model is effective in predicting delays with high accuracy. This accuracy provides airlines with the potential to develop systems that proactively manage delays before they occur. By implementing predictive models like this, airlines could send early warnings to passengers and make informed decisions about re-scheduling flights, reassigning gates, or reallocating crews to avoid cascading delays across flights.

From a business perspective, the results offer considerable value. By mitigating delays, airlines can reduce the substantial costs associated with fuel consump-

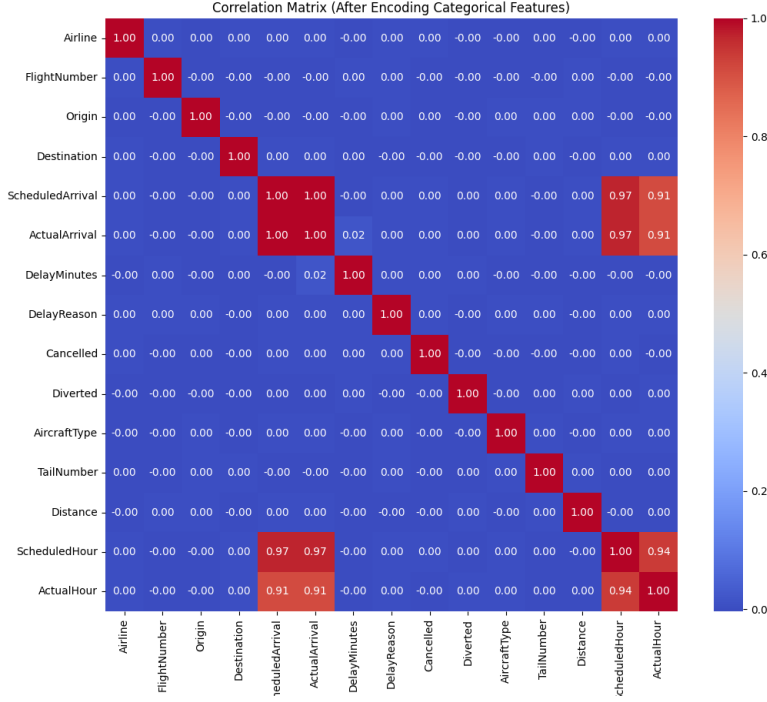


Fig. 3. Correlation Matrix (After Encoding Categorical Features)

tion, crew management, passenger compensation, and loss of customer loyalty. Considering that flight delays in India already affect a significant percentage of flights, improving prediction models could significantly enhance operational efficiency, reduce economic losses, and improve the passenger experience.

Furthermore, the broader aviation industry, including regulators such as the DGCA, could leverage this predictive model to set benchmarks for airline performance, monitor compliance with schedules, and encourage practices that reduce delays. The model could be integrated into nationwide delay prediction systems, helping airlines better manage resources and contribute to the overall efficiency of the aviation sector.

While this study has produced promising results, it is important to acknowledge its limitations. The model relies heavily on historical flight data, and any rapid changes in the aviation industry, such as new regulatory policies, unexpected weather patterns, or pandemic-related disruptions, could affect its predictive power. Future work should explore incorporating real-time data such as weather conditions, air traffic congestion, and aircraft maintenance records to enhance prediction accuracy.

In conclusion, the insights gained from this model offer airlines and aviation authorities a powerful tool to predict and manage flight delays. As data collection methods improve and the models become more sophisticated, we can expect predictive analytics to play an increasingly important role in minimizing delays and improving the overall efficiency of the aviation industry.

5 Threats to Validity

This section discusses potential limitations and sources of bias that may affect the reliability and generalizability of our study’s results.

5.1 Data Limitations

Historical Data Dependence: The model relies on historical flight data, which may not fully generalize to future scenarios if patterns shift due to changes in regulations, operational practices, or technology.

Incomplete Data: Missing or incomplete data, such as missing reasons for delays, could introduce bias or reduce the model’s predictive accuracy, especially if certain delay factors are underrepresented.

5.2 Generalizability

Regional Constraints: Since the model is trained on data specific to the Indian aviation context, it may have limited applicability to other regions with different regulatory frameworks, air traffic volumes, or weather patterns.

Flight Type Variation: The model may perform differently based on flight types (e.g., domestic vs. international) or specific airlines not well represented in the data, which could limit its broader applicability.

5.3 Model-Specific Bias

Algorithm Limitations: The Random Forest Regressor may introduce bias if certain features, such as origin or departure time, dominate predictions, potentially overlooking other relevant variables.

Feature Selection: Although selected features improve model performance, they may exclude relevant factors like real-time weather or air traffic conditions, limiting predictive accuracy.

5.4 External Factors

Unpredictable Events: Factors like extreme weather events, sudden regulatory changes, or global disruptions (e.g., pandemics) could impact flight delays in ways not captured by historical data, reducing the model’s robustness.

Seasonal Variation: Delay patterns may vary seasonally, and if the training data lacks seasonal balance, the model may exhibit performance variability.

5.5 Evaluation Metrics and Overfitting

Overfitting Risk: With a high R^2 score, the model may risk overfitting to the training data, potentially reducing its accuracy on new data. A robust validation process and alternative metrics could help mitigate this risk.

Metric Interpretation: High scores on metrics like Mean Squared Error (MSE) and R^2 could lead to a false sense of accuracy if the model lacks validation on diverse test cases representing real-world conditions.

By addressing these threats to validity, we aim to present a balanced view of the model's strengths and limitations, enhancing the credibility and practical relevance of our findings.

6 Related Work

In recent years, advanced methods have been developed to predict flight delays, incorporating both spatial and temporal factors. Li and Jing (2022) proposed a novel ST-Random Forest model that integrates complex network theory and LSTM units to capture the spatial and temporal dependencies affecting delays. Their model considers critical variables such as weather conditions, airport congestion, and aviation network topology, achieving a high classification accuracy of 92.39 in their case study on domestic flights in China. By analyzing spatial features at node, edge, and network levels, along with time-based factors like airport crowdedness and weather impacts, the study offers a robust prediction framework that aids real-time flight delay monitoring and provides insights for optimized air traffic management strategies.[3] This work complements other machine learning approaches, such as neural networks and ensemble models, which focus primarily on flight-specific attributes (e.g., departure times and airline). Compared to conventional methods that analyze flight delays at a macro level, this model's emphasis on the aviation network structure highlights the importance of micro-level spatial and temporal factors, contributing to more accurate delay forecasting.

Flight delays are a pervasive issue in the aviation industry, causing economic and operational challenges for airlines, airports, and passengers. To address this, Sternberg et al. (2020) provided a comprehensive review of methods for flight delay prediction [4] from a Data Science perspective. The authors propose a taxonomy to categorize existing approaches based on scope, data, and computational methods, highlighting the increasing reliance on machine learning techniques. They explore different problem aspects, such as delay propagation, root delay, and cancellation, each of which demands distinct predictive models. The review also identifies critical attributes—weather conditions, passenger demand, and operational schedules—that influence delay prediction accuracy. By mapping the evolution of delay prediction research, this paper outlines emerging trends and underscores the role of big data and machine learning in enhancing predictive accuracy, thus offering a roadmap for future research focused on improving operational efficiency in aviation.

7 Conclusion

This study aimed to enhance flight delay prediction in the Indian aviation sector by implementing a Random Forest Regressor model. By analyzing a comprehensive dataset of 524,288 flight records, we achieved an impressive predictive accuracy with an R^2 score of 0.9995, demonstrating the effectiveness of machine learning methods in addressing complex operational challenges within the aviation industry. Key variables influencing delays were identified, including scheduled departure times, origin airports, and specific airlines, which can inform strategic decision-making for airlines and airport management.

The implications of this research are significant, as accurate delay predictions can lead to improved operational efficiency, reduced costs, and enhanced passenger satisfaction. Airlines can leverage our model to proactively manage schedules, optimize resource allocation, and mitigate the cascading effects of delays throughout their networks. Additionally, stakeholders in the aviation sector can utilize these insights to enhance customer service and strengthen their competitive advantage.

Future research should focus on expanding the model's applicability to other geographical regions and incorporating real-time data streams, such as weather conditions and air traffic variations, to further refine predictions. Investigating the integration of advanced machine learning techniques, such as deep learning or ensemble methods, could also yield additional performance improvements. Furthermore, exploring the socio-economic impacts of flight delays and developing decision-support tools based on predictive analytics could provide valuable insights for stakeholders across the aviation ecosystem.

In conclusion, this study contributes to the growing body of literature on flight delay prediction and highlights the potential of data-driven approaches to address operational inefficiencies in the aviation industry, paving the way for a more resilient and responsive air transportation system.

References

1. Arikan, M.: The impact of airline flight schedules on flight delays: An analysis of block-time, delay propagation, and schedule optimization using stochastic models. Ph.D. thesis, Purdue University (2011)
2. Deshpande, V., Arikan, M.: The impact of airline flight schedules on flight delays. *Manufacturing & Service Operations Management* **14**(3), 423–440 (2012)
3. Li, Q., Jing, R.: Flight delay prediction from spatial and temporal perspective. *Expert Systems with Applications* **205**, 117662 (2022)
4. Sternberg, A., Soares, J., Carvalho, D., Ogasawara, E.: A review on flight delay prediction. arXiv preprint arXiv:1703.06118 (2017)